

2. LIMITATIONS

2.1 KINDS OF OPERATION

Approved for Commercial (COML) and Non-commercial (NON-COML) IFR operations.

NOTE Commercial IFR operation is only allowed when either SAS or AFCS is installed and operational, however, the applicable national operating regulations for IFR flights shall be complied with.

External load operations, installation of the Rescue Hoist (FMS 10-17), and flights with opened or removed doors are prohibited during IFR operations.

■ 2.2 Deleted

2.3 GROSS MASS

Model Variants	AFCS	SAS	Max IFR Gross Mass - kg	
			COML	NON-COML
CB, CB-4	X		2400	2400
CB, CB-4, CBS, CBS-4		X	2300	2400

2.4 LONGITUDINAL CENTER OF GRAVITY LIMITS

2.4.1 Commercial Operations

For Commercial IFR operations, the longitudinal CG limits as shown on the CG diagram (see Fig. 1) apply.

NOTE Cross-hatched area applies when the max IFR gross mass of 2400 kg is allowed (see para 2.3).

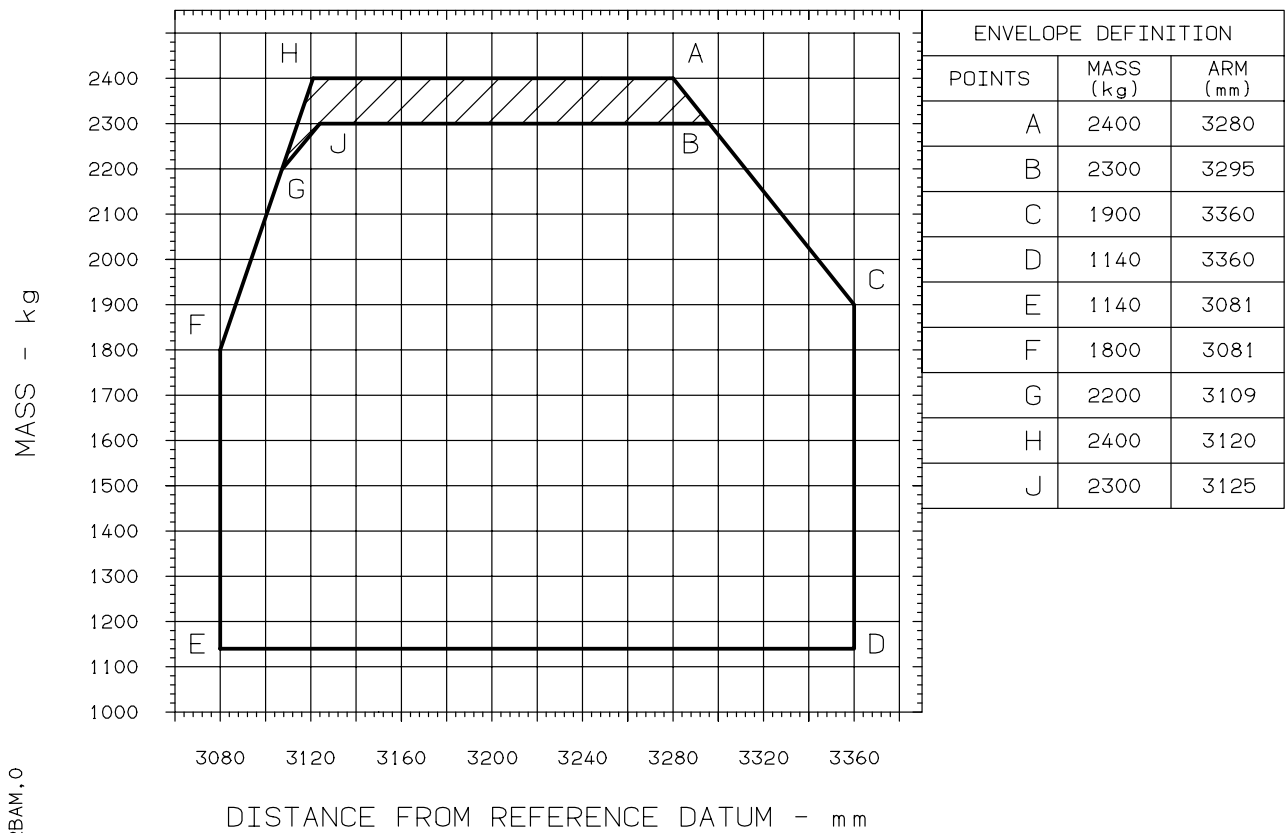


Fig. 1 Longitudinal Center of Gravity Limits

2.4.2 Non-commercial Operations

No change in the basic Flight Manual data.

2.5 AIRSPEED LIMITS

2.5.1 Maximum IFR Airspeed (V_{MAX})

2.5.1.1 Commercial Operations

BO 105 Model	AFCS	SAS	Max IFR Airspeed
C	X		$V_{NE} - 15 \text{ KIAS}^{1)}$ $V_{NE} - 40 \text{ KIAS} + \frac{\text{Alt (ft)}}{1000}^{2)}$
C		X	$V_{NE} - 20 \text{ KIAS}$
S		X	$V_{NE} - 15 \text{ KIAS}$
NOTE ¹⁾ Hands-on ²⁾ Hands-off			

2.5.1.2 Non-commercial Operations

Gross mass up to 2300 kg:

V_{MAX} $V_{NE} - 10\%$

Gross mass between 2300 and 2400 kg:

V_{MAX} $V_{NE} - 15 \text{ kt}$

2.5.2 Minimum IFR Airspeed (V_{MIN})

V_{MIN} (all variants) 45 KIAS

2.6 ALTITUDE LIMITS

2.6.1 IFR Flights with AFCS (FMS 10-41)

Gross mass up to 2300 kg:

Max altitude 17,000 ft

Gross mass between 2300 and 2400 kg:

Max altitude 10,000 ft

2.6.2 IFR Flights with Fuel Microfilters (FMS 10-39)

When fueled with MIL-T-5624, Grade JP-5; ASTM D-1655, Jet A and Jet A1; or equivalent:

Max altitude 14,200 ft

When fueled with MIL-T-5624, Grade JP-4; MIL-T-83133, Grade JP-8; ASTM D-1655, Jet B; Alternate Fuel (AVGAS/Jet Fuel Mix); or equivalent:

Max altitude 8,000 ft

2.7 AMBIENT AIR LIMITS

2.7.1 Outside Air Temperature Limits

Operation of the SAS (FMS 10-29) is prohibited when OAT is less than -35°C

2.7.2 Turbulent Air Limits

Intentional flight into areas of severe turbulence is not permissible.

2.8 ROTOR RPM (N_{RO}) LIMITS

When operating the AFCS (FMS 10-41) observe the following permissible rotor RPM ranges:

Operating Condition	Permissible N_{RO} Range
Hands-on	98 - 102 % (100 % recommended)
Hands-off	100 - 102 %

2.9 PLACARDS AND DECALS

EFFECTIVITY Model C Variants with SAS (FMS 10-29)

Text:

IFR	$V_{MAX} = V_{NE} - 20 \text{ KT}$
	$V_{MIN} = 45 \text{ KT}$

Location: Adjacent V_{NE} -table placard, in pilot's field-of-view.

EFFECTIVITY Model S Variants with SAS (FMS 10-29)

Text:

IFR	$V_{MAX} = V_{NE} - 15 \text{ KT}$
	$V_{MIN} = 45 \text{ KT}$

Location: Adjacent V_{NE} -table placard, in pilot's field-of-view.

EFFECTIVITY Model C Variants with AFCS (FMS 10-41)

Text:

SAS SPERRY HELIPLOT	MAX ALT UP TO 2300 KG = 17000 FT; ABOVE 2300 UP TO 2400 KG = 10000 FT		
	VFR-HANDS-ON	IFR-HANDS-ON	VFR/IFR-HANDS-OFF
V_{MAX}	V_{NE}	$V_{NE} - 15 \text{ KT}$	$V_{NE} - 40 \text{ KT} + \frac{\text{ALT (FT)}}{1000}$
V_{MIN}	_____	45 KT	45 KT
n_{RO}	98 - 102 %	98 - 102 %	100 - 102 %
NOTE HANDS-OFF-OPERATION ONLY IN ATTITUDE HOLD WITH HP1, HP2 AND YAW ENGAGED			

Location: Cockpit interior, in pilot's field-of-view